DEEP LEARNING · COMPUTER VISION · FINANCIAL TIME SERIES · 2005-2024

Market Regime Detection via CNN on Rolling Cross-Asset Correlation Heatmaps

9 assets · 30-day rolling Pearson correlation matrix → 64×64 PNG → Custom CNN classifier

0.46	61%	102k	4,878
MACRO-F1	BEAR RECALL	PARAMS	HEATMAPS

TensorFlow/Keras · yfinance · seaborn · Weights & Biases · Kaggle T4 GPU · Grad-CAM

MOTIVATION

During market stress, cross-asset correlations **collapse toward +1** — equities move together while bonds and gold decouple. This structural shift is **invisible to price-level models** but visible as a spatial pattern in the correlation matrix.

We render 30-day rolling Pearson matrices as **64×64 heatmap images** and train a CNN to classify regime from pixel structure alone — a pure vision approach to a finance problem.

ASSET UNIVERSE — 9 ASSETS

EQUITY
XLF · XLK · XLE
XLV · XLI · VNQ

MACRO
GLD · TLT
VIX (level only)

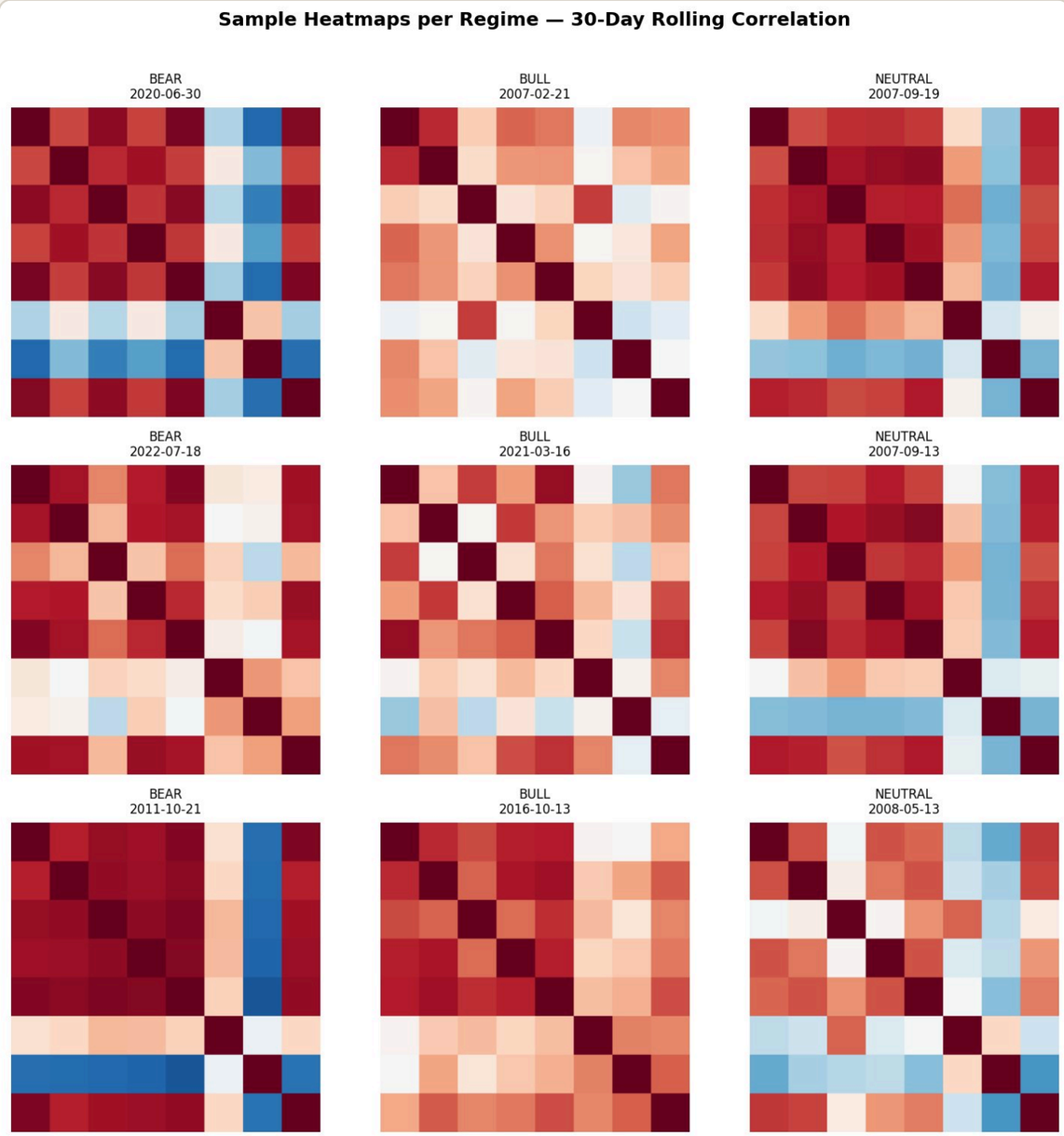
DATA PREPARATION & LABELS

Regime	Rule
Bear	`VIX>30` OR `dd>15%`
Bull	`VIX<20` AND `dd<5%`
Neutral	Residual

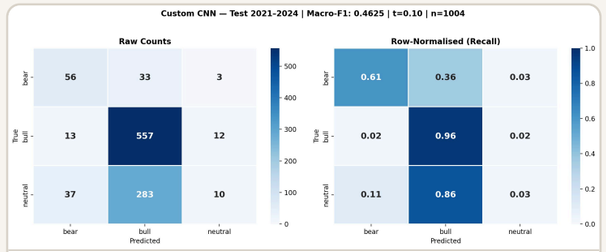
SPY 126-day rolling drawdown. Bear OR = many failure modes captured. Bull AND = strict. Chronological split: train 2005–18 · val 2019–20 · test 2021–24.

● Bear 483 ● Bull 3,028 ● Neutral 1,396

INPUT REPRESENTATION — 30-DAY CORRELATION HEATMAPS



CONFUSION MATRIX (T = 0.10, N = 1,004)



Bear recall 61% (56/92). Bull recall 96%. Neutral recall 3% — model treats neutral as residual, defaulting uncertainty to bull.

RESULTS — TEST SET (T = 0.10)

Class	Prec.	Rec.	F1
Bear	0.528	0.609	0.566
Bull	0.638	0.957	0.766
Neutral	0.400	0.030	0.056
Macro avg	**0.522**	**0.532**	**0.463**

Model	Val F1	Test F1
CNN argmax	0.643	0.414
CNN t=0.10 ★	**—**	**0.463**
MobileNetV2	0.605	~0.32

102k-param domain-specific CNN outperforms 2.3M-param ImageNet backbone

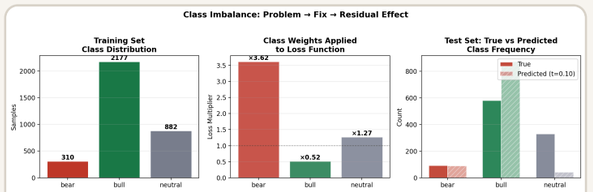
THRESHOLD ANALYSIS — PER-CLASS F1 TRADE-OFF

CNN ARCHITECTURE

```
Input 64×64×3                                   PNG heatmap
↓
Conv2D(32) + BN      MaxPool → 32²
Conv2D(64) + BN      MaxPool → 16²
Conv2D(128) + BN     MaxPool → 8²
GlobalAvgPool → 128                             no Flatten
Dense(64) + Dropout(0.4)
Softmax(3)                                      bear / bull / neutral
```

GAP prevents 524k head param explosion on ~3,400 train samples. BN stabilises training depth. Dropout(0.4) regularises the dense head only.

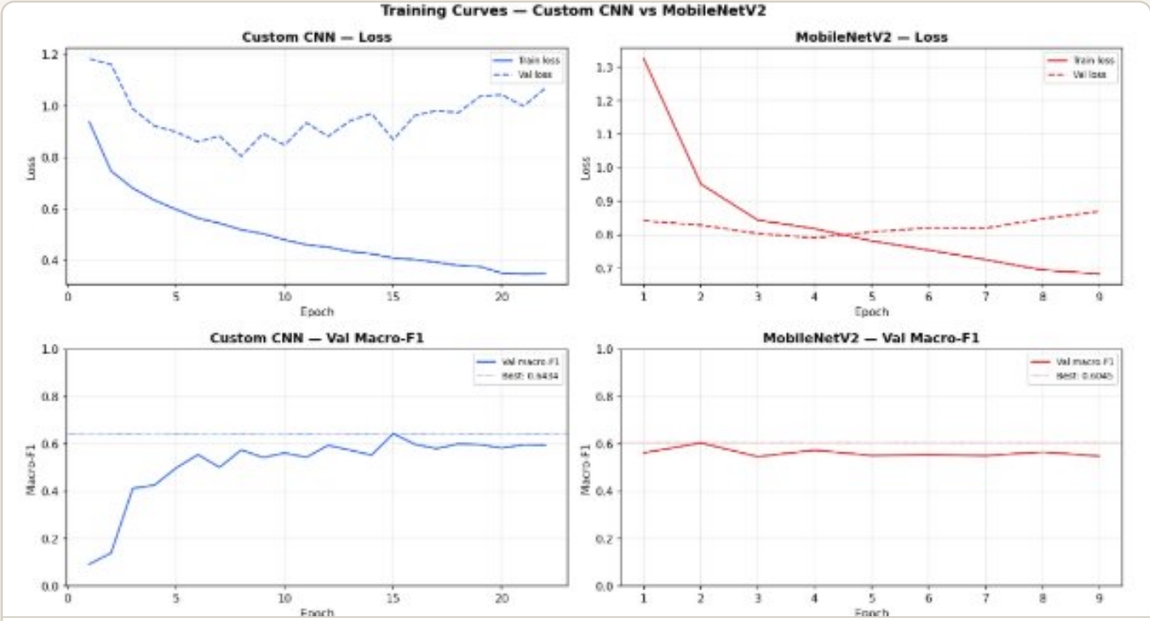
CLASS IMBALANCE: PROBLEM → FIX → EFFECT



Training set is 65% bull. Class weights multiply bear loss by ×3.62, forcing the network to learn the rare class. Right panel shows the residual effect: model over-predicts bull on test set despite the fix.

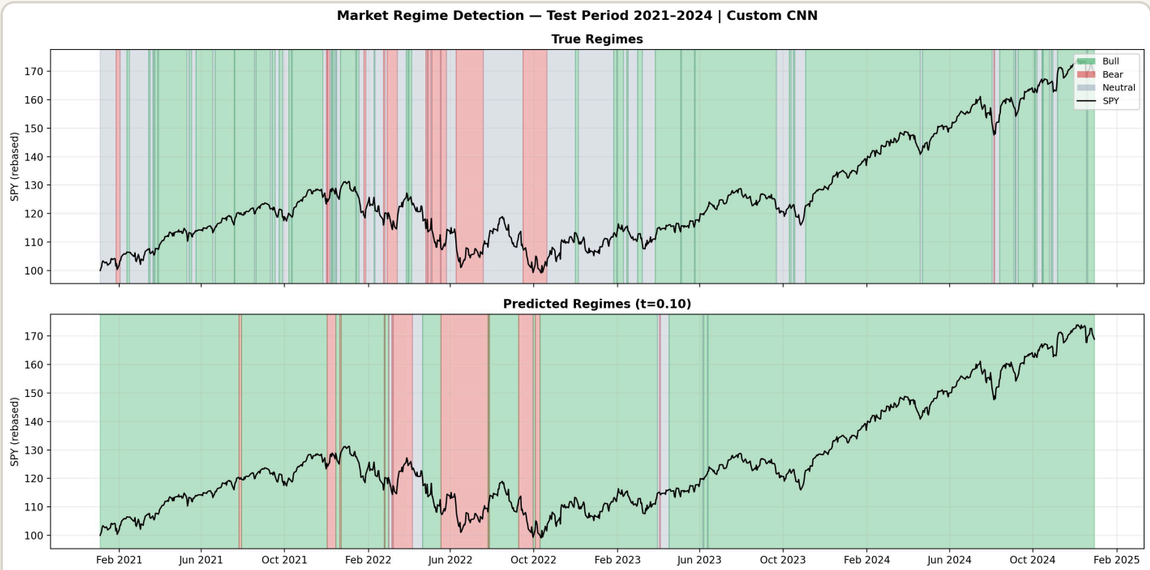
Bear (left): uniform red equity block + blue TLT/GLD columns — crisis correlation spike. Bull (centre): muted, mixed structure. Neutral (right): intermediate. Fixed vmin=-1, vmax=+1 encodes absolute correlation magnitude as pixel intensity.

TRAINING CURVES — CUSTOM CNN VS MOBILENETV2

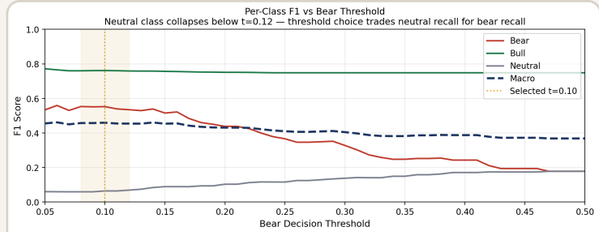


Custom CNN peaks val macro-F1=0.64 at epoch 15, early-stopped at epoch 22. MobileNetV2 (frozen ImageNet backbone) plateaus at epoch 2 — photographic priors do not transfer to numerical color-encoded correlation matrices.

REGIME TIMELINE — TEST PERIOD 2021-2024



Top: true regimes shaded over rebased SPY. Bottom: predictions at t=0.10. Model correctly captures the 2022 bear cluster (Jun–Oct) and most 2021 bear days, but neutral is near-absent — threshold lowers the decision boundary for bear at neutral's expense.

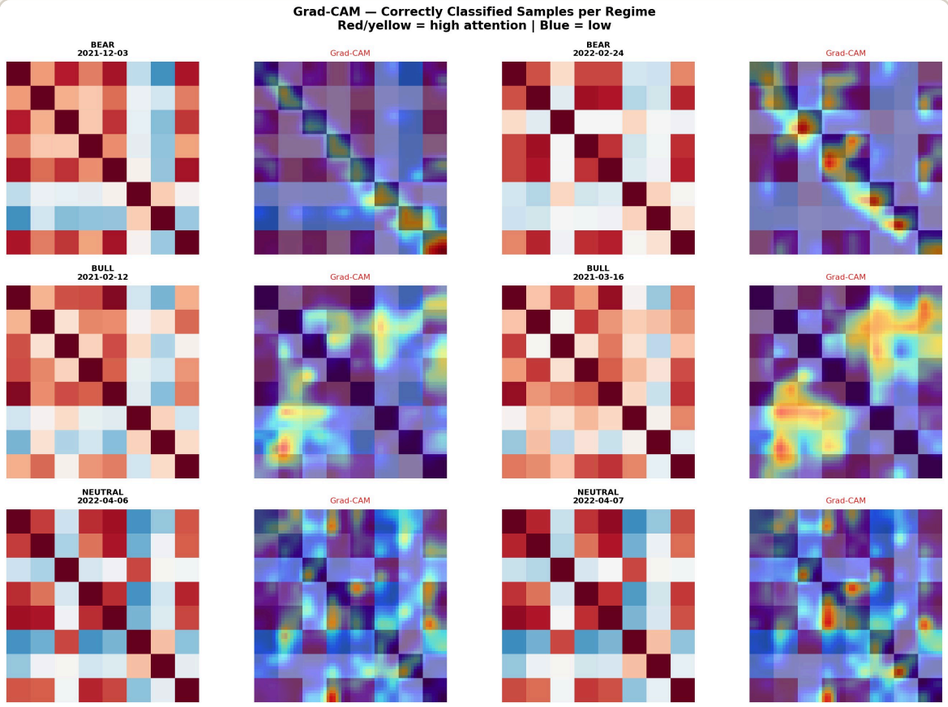


Bear F1 maximised at low t; neutral F1 collapses below t=0.12. t=0.10 is the operating point that maximises bear recall while macro-F1 remains above the argmax baseline.

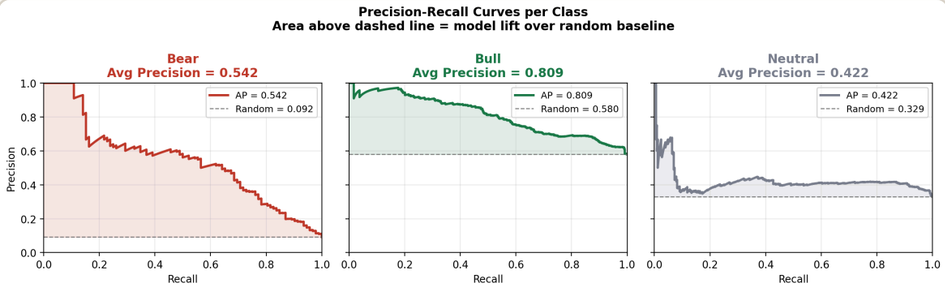
KEY FINDING

Threshold calibration recovered **51 pts of bear recall without retraining** — revealing a genuine but miscalibrated signal. The network learned crisis bears well; the 2022 slow-grind bear required a lower decision boundary to surface.

MODEL INTERPRETATION — GRAD-CAM & PRECISION-RECALL CURVES



Correctly-classified samples per regime. Bear — model attends to off-diagonal cells where TLT/GLD decouple from equities, confirming it learned the genuine crisis signature. Bull — diffuse attention. Neutral — low-magnitude scattered attention, consistent with no sharp visual signature.



PR curves vs random baseline. Bear AP=0.542 vs random=0.092 — 5.9× lift. Bull AP=0.809. Neutral AP=0.422 barely above random (0.329), confirming weak separability in correlation space. All classes show genuine learned signal.

KEY FINDINGS

9.8% → 61%
Bear recall via threshold calibration at t=0.10 — no retraining required

5.9× lift
Bear AP over **random baseline** — CNN genuinely learned crisis correlation structure

+14 F1
Over **MobileNetV2** — ImageNet priors don't transfer to numerical color encodings

3% recall
Neutral class — no sharp visual signature; AP barely above random at 0.422

WHY CNN FOR THIS TASK?

Correlation matrices are **structured 2D inputs** — the equity block occupies top-left, bond/gold columns bottom-right. A CNN's inductive bias (local filters, translation equivariance) directly matches this: a 3×3 filter learns a local correlation transition; stacked layers compose these into block-level regime patterns.

Alternatives (MLP, tree methods) require flattening the matrix, losing the spatial structure that carries the regime signal. Grad-CAM confirms the network exploits this structure — attending to specific off-diagonal regions rather than the uninformative diagonal.

LIMITATIONS & FUTURE WORK

Neutral class: residual labelling gives it no clean visual identity. A richer labelling scheme (late-bull / early-bear sub-regimes) would likely improve recall significantly.

Test-period generalisation: model trained on COVID-style crises (val 2019–20); 2022 slow grind required threshold recalibration. A multi-period val set would produce better-calibrated probabilities.

Future: eigenvalue decomposition as side-channel input · temporal smoothing prior · fine-tuning MobileNetV2 on domain data.

REPRODUCIBILITY

`RUN_FULL=False` — demo mode, ~30 sec on T4

`RUN_FULL=True` — full pipeline, ~25 min on T4

Companion: `market_regime_cnn.ipynb`